

NeuroLens

Private, real-time AI for cognitive accessibility

Understand the moment. Organize the thought. Keep everything private.

Fast, indirect communication hides the real task

What was said

“It would probably be good if you could get the analysis over to me sometime before Friday.”

What a person has to work out, live

Is this a request or a suggestion?

What is the deadline, exactly?

What else is unclear?

1

Indirect wording

Polite phrasing can hide real requests and deadlines.

2

No time to parse

Lectures and meetings move on before you have worked it out.

3

Cloud help costs privacy

Assistants that fix this usually need your conversations sent to a remote server.

What is missing today

Transcription only

Gives you every word.

Gap

Does not say what matters or what to do.

Cloud AI assistants

Can summarize and answer.

Gap

Needs private conversations sent off-device.

Task apps

Organize tasks well.

Gap

Start from tasks you type in yourself.

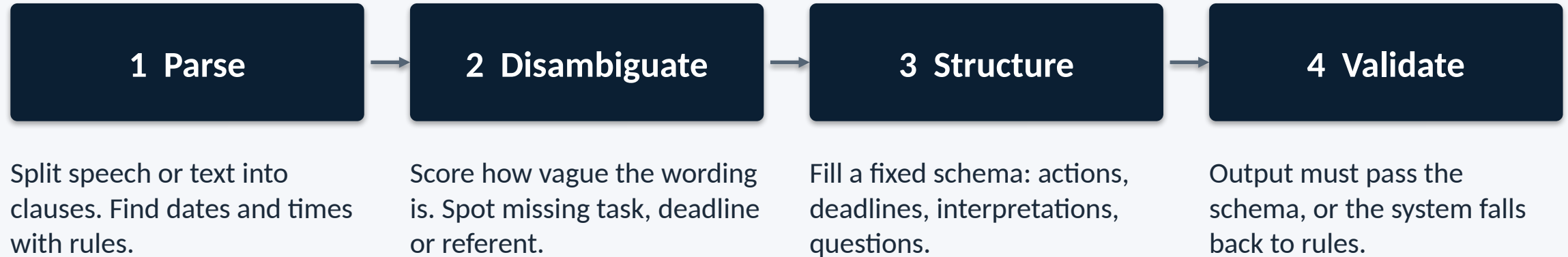
Overconfident answers

Sound certain.

Gap

Rarely say what is unclear or ask instead.

From messy language to structure



Not a chatbot. It returns the same fixed structure every time, so every field can be shown, checked and reused.

Known ≠ Inferred ≠ Unknown

KNOWN

What is stated

Literal meaning, explicit tasks, deadlines found by rules.

From the meeting example

Deadline: Friday

INFERRED

What is possible

Readings of vague wording, always with the words that support them.

From the meeting example

May be a real request, not just a suggestion

UNKNOWN

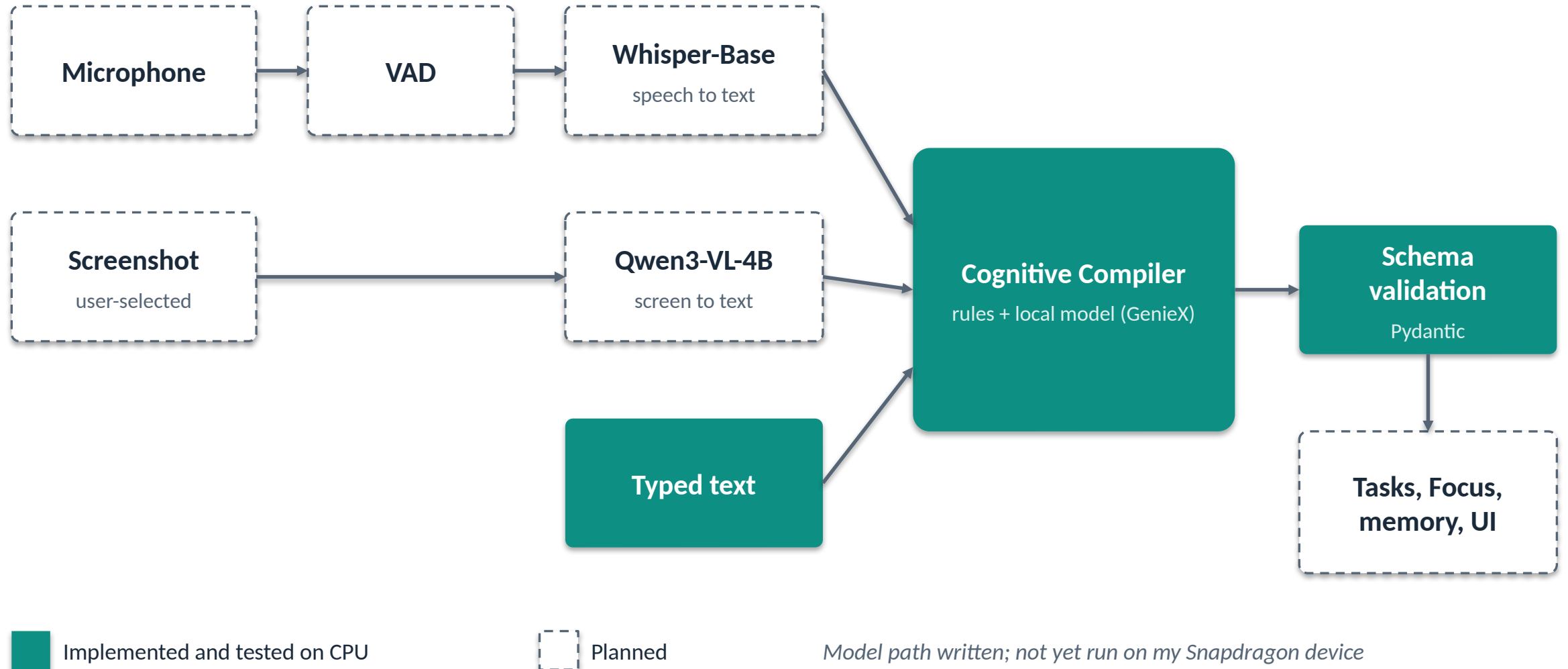
What is missing

Gaps become a question, not a guess.

From the meeting example

“Which part should be changed, and how?”

Architecture: what exists and what is planned



Snapdragon plan: targets, not results

Role	Candidate model	Intended runtime	Status
Speech to text	Whisper-Base	Qualcomm AI Hub / QAI AppBuilder	Not yet run on my device
Reasoning	Qwen3-0.6B (Phi-4-Mini alt.)	GenieX local server	Client written; not yet run
Screen understanding	Qwen3-VL-4B-Instruct	GenieX	Planned

Inference is intended to run on the Snapdragon NPU where supported; UI, storage and rules run on the CPU. **No speed or accuracy numbers are claimed until measured on my own laptop.**

Privacy by design (goals for the full app)

1

Ephemeral by default

Audio and screenshots are processed in memory and discarded unless the user chooses to save.

2

Preferences, not logs

Memory keeps user-approved settings such as “prefers short answers”, not conversation history.

3

No cloud AI for core use

Local models handle inference. Offline behaviour will be tested before it is claimed.

These are design goals; the privacy features are not built yet.

Prototype today: real output from the rule-based engine

Input

“It would probably be good if you could get the analysis over to me sometime before Friday. And maybe revisit the first section because I don't think we're quite there yet.”

Output

Actions

- [] get the analysis over to me (high)
- [] revisit the first section (medium)

Deadline

Friday

Ambiguity

MEDIUM

Clarifying question

“Which part should be changed, and how?”

Produced on CPU by the rule engine (python cognitive_compiler.py --no-model). The local-model path has not yet been run on Snapdragon.

Where it stands and what comes next

Done

- Structured-output schema
- Rule-based compiler
- Model client with validation and fallback
- 16 unit tests

Next

- Run GenieX + Qwen3 on my laptop
- Whisper-Base on audio
- Task creation, Focus Mode, simple UI
- Screenshot input with Qwen3-VL

To measure

- Task and deadline accuracy on ~30 labelled samples
- Latency: ASR, first token, end to end
- CPU vs NPU
- Offline test